

Box 0 The Broken Spreadsheet

An AI lab tracks every training run in one shared CSV file.

model	researcher	email	lab	dataset	sz_tb	license
GPT-X	Alice Chen	alice@lab.ai	VisionLab	ImageNet-22k	1.2	CC-BY
GPT-X	Alice Chen	alice@lab.ai	VisionLab	LAION-5B	5.0	CC-BY
LLaMA-4	Alice Chen	alice@lab.ai	NLPLab	C4	0.8	ODC
LLaMA-4	Bob Nkosi	bob@lab.ai	NLPLab	C4	0.8	ODC
DiffNet	Carol Diaz	carol@lab.ai	VisionLab	COCO-2024	0.3	CC-BY-NC

Your task (8 minutes):

1. Circle every piece of **redundant data** in the table.
2. Sketch **two or more relations** that fix it. Name each one, list its attributes, and underline the primary key.
3. Draw an **arrow** wherever a foreign key exists.

Sketch your relations here.

Box 1 Chen ERD Sketch (4 minutes)

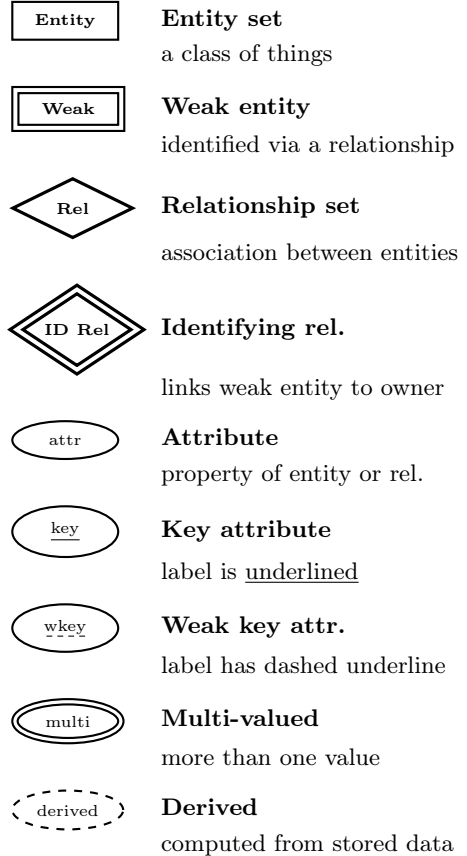
You have **Researcher** and **Lab** from the lecture. Add a third entity: **Project**.

- A Researcher works on **many** Projects.
- A Project has **exactly one** Lead Researcher.
- Show: entity rectangle, at least two attributes (one key, underlined), relationship diamond, cardinality.

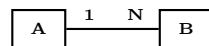
Sketch your ERD here.

Debrief: what did you underline as the key for Project? Why not the title?

Chen ER Notation



Cardinality



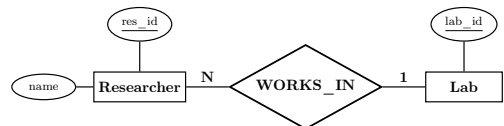
1 = exactly one N/M = many

1:1 one-to-one

1:N one-to-many

M:N many-to-many

Example



A Researcher WORKS_IN one Lab. A Lab has N Researchers.

GenAI Reflection

Scenario. You have just been hired as a junior database engineer at an AI lab. Your first task: design the schema for tracking all training runs – models, datasets, researchers, compute costs, water usage per run. Your tech lead says: “*Use whatever tools you want.*”

1. Where in your workflow would you use an LLM?

Think concretely, which specific tasks would you hand off? (e.g. boilerplate SQL, explaining an error, generating a first draft.)

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2. Where would you explicitly *not* trust it, and why?

Name at least two specific design decisions. Explain what the LLM is missing that you would need to supply.

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3. The water usage column.

A colleague says: “*Just ask ChatGPT to write the `CREATE TABLE`, it’ll get it right.*”

(a) What will it probably get right?

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(b) What will it almost certainly get wrong or invent, and why?

Hint: what information would it need that isn’t in your prompt?

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Notes:

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